

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Vikram | Full-Stack Developer Portfolio</title>
<meta name="description" content="Vikram - BCA student passionate about full-stack web
development, cloud computing, and AI. Explore my projects and skills.">
<meta name="keywords" content="Vikram, full-stack developer, Python, Django, SQL, web
development, cloud computing, portfolio">
<meta name="author" content="Vikram">

<!-- Open Graph Tags -->
<meta property="og:title" content="Vikram | Full-Stack Developer Portfolio">
<meta property="og:description" content="BCA student passionate about full-stack web
development, cloud computing, and AI.">
<meta property="og:type" content="website">
<meta property="og:url" content="https://vikram-portfolio.com">
<meta property="og:image" content="https://vikram-portfolio.com/images/og-image.jpg">

<!-- JSON-LD Schema -->
<script type="application/ld+json">
{
"@context": "https://schema.org",
"@type": "Person",
"name": "Vikram",
"url": "https://vikram-portfolio.com",
"description": "BCA student passionate about full-stack web development, cloud computing,
and AI.",
"knowsAbout": ["Python", "Django", "SQL", "Full-Stack Development", "Cloud Computing",
"AI"],
"alumniOf": {
"@type": "EducationalOrganization",
"name": "University Name"
}
}
</script>

<!-- Preload Resources -->
<link rel="preload"
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&family=Orbitron:wght@
as="style">
```

```
<link rel="preload" href="https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js"
as="script">
```

```
<!-- Fonts -->
```

```
<link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&family=Orbitron:wght@400;500;600;700&display=block"
rel="stylesheet">
```

```
<!-- AOS Library -->
```

```
<link href="https://unpkg.com/aos@2.3.1/dist/aos.css" rel="stylesheet">
```

```
<style>
```

```
:root {
  --primary: #0a0a12;
  --secondary: #1a1a2e;
  --accent-cyan: #00f5ff;
  --accent-purple: #8a2be2;
  --accent-green: #00ff9d;
  --text-primary: #ffffff;
  --text-secondary: #b0b0b0;
  --glass-bg: rgba(255, 255, 255, 0.05);
  --glass-border: rgba(255, 255, 255, 0.1);
  --shadow-glow: 0 0 20px rgba(0, 245, 255, 0.3);
  --shadow-glow-purple: 0 0 20px rgba(138, 43, 226, 0.3);
}
```

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
```

```
html {
  scroll-behavior: smooth;
}
```

```
body {
  font-family: 'Inter', sans-serif;
  background: var(--primary);
  color: var(--text-primary);
  overflow-x: hidden;
  line-height: 1.6;
```

```
}
```

```
h1, h2, h3, h4, h5 {  
font-family: 'Orbitron', sans-serif;  
font-weight: 700;  
margin-bottom: 1rem;  
}
```

```
h1 {  
font-size: 3.5rem;  
background: linear-gradient(90deg, var(--accent-cyan), var(--accent-purple));  
-webkit-background-clip: text;  
background-clip: text;  
color: transparent;  
margin-bottom: 1.5rem;  
}
```

```
h2 {  
font-size: 2.5rem;  
color: var(--accent-cyan);  
position: relative;  
display: inline-block;  
margin-bottom: 2rem;  
}
```

```
h2::after {  
content: " ";  
position: absolute;  
bottom: -10px;  
left: 0;  
width: 60px;  
height: 3px;  
background: var(--accent-cyan);  
}
```

```
p {  
margin-bottom: 1.5rem;  
color: var(--text-secondary);  
}
```

```
.container {  
width: 90%;
```

```
max-width: 1200px;
margin: 0 auto;
padding: 0 15px;
}
```

/ Navigation /

```
nav {
position: fixed;
top: 0;
left: 0;
width: 100%;
z-index: 1000;
padding: 1.5rem 0;
transition: all 0.3s ease;
backdrop-filter: blur(10px);
background: rgba(10, 10, 18, 0.8);
border-bottom: 1px solid var(--glass-border);
}
```

```
nav.scrolled {
padding: 1rem 0;
background: rgba(10, 10, 18, 0.95);
box-shadow: var(--shadow-glow);
}
```

```
.nav-container {
display: flex;
justify-content: space-between;
align-items: center;
}
```

```
.logo {
font-family: 'Orbitron', sans-serif;
font-size: 1.8rem;
font-weight: 700;
color: var(--accent-cyan);
text-decoration: none;
}
```

```
.nav-links {
display: flex;
list-style: none;
```

```
}
```

```
.nav-links li {  
margin-left: 2rem;  
}
```

```
.nav-links a {  
color: var(--text-primary);  
text-decoration: none;  
font-weight: 500;  
transition: all 0.3s ease;  
position: relative;  
}
```

```
.nav-links a:hover {  
color: var(--accent-cyan);  
}
```

```
.nav-links a::after {  
content: "";  
position: absolute;  
bottom: -5px;  
left: 0;  
width: 0;  
height: 2px;  
background: var(--accent-cyan);  
transition: width 0.3s ease;  
}
```

```
.nav-links a:hover::after {  
width: 100%;  
}
```

```
.mobile-menu-btn {  
display: none;  
background: none;  
border: none;  
color: var(--text-primary);  
font-size: 1.5rem;  
cursor: pointer;  
}
```

/ Hero Section /

```
.hero {  
  height: 100vh;  
  position: relative;  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  overflow: hidden;  
}
```

```
#three-canvas {  
  position: absolute;  
  top: 0;  
  left: 0;  
  width: 100%;  
  height: 100%;  
  z-index: -1;  
}
```

```
.hero-content {  
  text-align: center;  
  z-index: 1;  
  max-width: 800px;  
  padding: 0 20px;  
}
```

```
.hero h1 {  
  font-size: 4rem;  
  margin-bottom: 1rem;  
  text-shadow: 0 0 10px rgba(0, 245, 255, 0.5);  
}
```

```
.hero p {  
  font-size: 1.2rem;  
  margin-bottom: 2rem;  
  color: var(--text-secondary);  
}
```

```
.cta-button {  
  display: inline-block;  
  padding: 12px 30px;  
  background: linear-gradient(45deg, var(--accent-cyan), var(--accent-purple));  
}
```

```
color: var(--text-primary);
text-decoration: none;
border-radius: 30px;
font-weight: 600;
transition: all 0.3s ease;
box-shadow: var(--shadow-glow);
border: none;
cursor: pointer;
font-size: 1rem;
}

.cta-button:hover {
transform: translateY(-3px);
box-shadow: 0 10px 20px rgba(0, 245, 255, 0.4);
}
```

```
/ Sections /
section {
padding: 100px 0;
}
```

```
.section-title {
text-align: center;
margin-bottom: 60px;
}
```

```
/ About Section /
.about-content {
display: grid;
grid-template-columns: 1fr 1fr;
gap: 50px;
align-items: center;
}
```

```
.about-text {
padding-right: 20px;
}
```

```
.about-image {
position: relative;
border-radius: 10px;
overflow: hidden;
```

```
box-shadow: var(--shadow-glow);
}
```

```
.about-image::before {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: linear-gradient(45deg, var(--accent-cyan), var(--accent-purple));
  opacity: 0.3;
  z-index: 1;
}
```

```
.about-image img {
  width: 100%;
  height: auto;
  display: block;
  transition: transform 0.5s ease;
}
```

```
.about-image:hover img {
  transform: scale(1.05);
}
```

/ Skills Section /

```
.skills-container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 30px;
}
```

```
.skill-card {
  background: var(--glass-bg);
  backdrop-filter: blur(10px);
  border: 1px solid var(--glass-border);
  border-radius: 10px;
  padding: 30px;
  text-align: center;
  transition: all 0.3s ease;
  box-shadow: 0 5px 15px rgba(0, 0, 0, 0.1);
}
```



```
}
```

```
.skill-card:hover {  
transform: translateY(-10px);  
box-shadow: var(--shadow-glow);  
border-color: var(--accent-cyan);  
}
```

```
.skill-icon {  
font-size: 2.5rem;  
margin-bottom: 15px;  
color: var(--accent-cyan);  
}
```

/ Projects Section /

```
.projects-container {  
display: grid;  
grid-template-columns: repeat(auto-fill, minmax(350px, 1fr));  
gap: 30px;  
}
```

```
.project-card {  
background: var(--glass-bg);  
backdrop-filter: blur(10px);  
border: 1px solid var(--glass-border);  
border-radius: 10px;  
overflow: hidden;  
transition: all 0.3s ease;  
box-shadow: 0 5px 15px rgba(0, 0, 0, 0.1);  
}
```

```
.project-card:hover {  
transform: translateY(-10px);  
box-shadow: var(--shadow-glow);  
border-color: var(--accent-cyan);  
}
```

```
.project-image {  
height: 200px;  
overflow: hidden;  
}
```

```
.project-image img {  
width: 100%;  
height: 100%;  
object-fit: cover;  
transition: transform 0.5s ease;  
}
```

```
.project-card:hover .project-image img {  
transform: scale(1.1);  
}
```

```
.project-content {  
padding: 20px;  
}
```

```
.project-title {  
font-size: 1.3rem;  
margin-bottom: 10px;  
color: var(--accent-cyan);  
}
```

```
.project-tech {  
display: flex;  
flex-wrap: wrap;  
gap: 10px;  
margin-bottom: 15px;  
}
```

```
.tech-tag {  
background: rgba(0, 245, 255, 0.1);  
color: var(--accent-cyan);  
padding: 5px 10px;  
border-radius: 20px;  
font-size: 0.8rem;  
}
```

/ Controls Section /

```
.controls-section {  
background: var(--secondary);  
border-radius: 15px;  
padding: 40px;  
margin-top: 50px;
```

```
box-shadow: var(--shadow-glow-purple);
}
```

```
.control-group {
display: flex;
flex-wrap: wrap;
gap: 30px;
margin-bottom: 30px;
}
```

```
.control-item {
flex: 1;
min-width: 250px;
}
```

```
.control-label {
display: block;
margin-bottom: 10px;
color: var(--accent-cyan);
font-weight: 600;
}
```

```
.slider-container {
display: flex;
align-items: center;
gap: 15px;
}
```

```
.slider {
flex: 1;
-webkit-appearance: none;
height: 8px;
background: rgba(255, 255, 255, 0.1);
border-radius: 5px;
outline: none;
}
```

```
.slider::-webkit-slider-thumb {
-webkit-appearance: none;
width: 20px;
height: 20px;
border-radius: 50%;
```

```
background: var(--accent-cyan);
cursor: pointer;
box-shadow: var(--shadow-glow);
}
```

```
.slider-value {
min-width: 40px;
text-align: center;
font-weight: 600;
color: var(--accent-cyan);
}
```

/ Contact Section /

```
.contact-container {
display: grid;
grid-template-columns: 1fr 1fr;
gap: 50px;
}
```

```
.contact-info {
display: flex;
flex-direction: column;
gap: 20px;
}
```

```
.contact-item {
display: flex;
align-items: center;
gap: 15px;
}
```

```
.contact-icon {
font-size: 1.5rem;
color: var(--accent-cyan);
}
```

```
.contact-form {
background: var(--glass-bg);
backdrop-filter: blur(10px);
border: 1px solid var(--glass-border);
border-radius: 10px;
padding: 30px;
```

```
}
```

```
.form-group {  
margin-bottom: 20px;  
}
```

```
.form-label {  
display: block;  
margin-bottom: 8px;  
color: var(--text-primary);  
font-weight: 500;  
}
```

```
.form-input, .form-textarea {  
width: 100%;  
padding: 12px 15px;  
background: rgba(255, 255, 255, 0.05);  
border: 1px solid var(--glass-border);  
border-radius: 5px;  
color: var(--text-primary);  
font-family: 'Inter', sans-serif;  
transition: all 0.3s ease;  
}
```

```
.form-input:focus, .form-textarea:focus {  
outline: none;  
border-color: var(--accent-cyan);  
box-shadow: 0 0 0 2px rgba(0, 245, 255, 0.2);  
}
```

```
.form-textarea {  
min-height: 150px;  
resize: vertical;  
}
```

/ Ad Placeholders /

```
.ad-container {  
background: var(--glass-bg);  
border: 1px dashed var(--glass-border);  
border-radius: 5px;  
padding: 20px;  
text-align: center;
```

```
margin: 30px 0;
color: var(--text-secondary);
}
```

```
.ad-sidebar {
position: fixed;
right: 20px;
top: 50%;
transform: translateY(-50%);
width: 160px;
background: var(--glass-bg);
border: 1px dashed var(--glass-border);
border-radius: 5px;
padding: 15px;
text-align: center;
color: var(--text-secondary);
z-index: 100;
}
```

/ Footer /

```
footer {
background: var(--secondary);
padding: 60px 0 30px;
text-align: center;
}
```

```
.footer-content {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
gap: 40px;
margin-bottom: 40px;
}
```

```
.footer-column h3 {
color: var(--accent-cyan);
margin-bottom: 20px;
font-size: 1.3rem;
}
```

```
.footer-links {
list-style: none;
}
```

```
.footer-links li {  
margin-bottom: 10px;  
}
```

```
.footer-links a {  
color: var(--text-secondary);  
text-decoration: none;  
transition: color 0.3s ease;  
}
```

```
.footer-links a:hover {  
color: var(--accent-cyan);  
}
```

```
.social-links {  
display: flex;  
justify-content: center;  
gap: 20px;  
margin-top: 20px;  
}
```

```
.social-link {  
display: inline-flex;  
align-items: center;  
justify-content: center;  
width: 40px;  
height: 40px;  
background: var(--glass-bg);  
border-radius: 50%;  
color: var(--text-primary);  
text-decoration: none;  
transition: all 0.3s ease;  
}
```

```
.social-link:hover {  
background: var(--accent-cyan);  
transform: translateY(-3px);  
box-shadow: var(--shadow-glow);  
}
```

```
.copyright {
```

```
padding-top: 30px;
border-top: 1px solid var(--glass-border);
color: var(--text-secondary);
font-size: 0.9rem;
}
```

/ Responsive Design /

```
@media (max-width: 992px) {
h1 {
font-size: 3rem;
}
```

```
h2 {
font-size: 2rem;
}
```

```
.about-content, .contact-container {
grid-template-columns: 1fr;
}
```

```
.ad-sidebar {
display: none;
}
}
```

```
@media (max-width: 768px) {
.mobile-menu-btn {
display: block;
}
```

```
.nav-links {
position: fixed;
top: 70px;
left: -100%;
width: 100%;
height: calc(100vh - 70px);
background: var(--primary);
flex-direction: column;
align-items: center;
justify-content: flex-start;
padding-top: 50px;
transition: left 0.3s ease;
```



```
}
```

```
.nav-links.active {  
left: 0;  
}
```

```
.nav-links li {  
margin: 15px 0;  
}
```

```
.hero h1 {  
font-size: 2.5rem;  
}
```

```
.projects-container {  
grid-template-columns: 1fr;  
}  
}
```

```
@media (max-width: 576px) {  
section {  
padding: 60px 0;  
}
```

```
.controls-section {  
padding: 20px;  
}
```

```
.control-group {  
flex-direction: column;  
}  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- Navigation -->
```

```
<nav id="navbar">
```

```
<div class="container nav-container">
```

```
<a href="#" class="logo">Vikram</a>
```

```
<button class="mobile-menu-btn" id="mobileMenuBtn">&0</button>
```

```
<ul class="nav-links" id="navLinks">
```

```
<li><a href="#home">Home</a></li>
```

```
<li><a href="#about">About</a></li>
<li><a href="#skills">Skills</a></li>
<li><a href="#projects">Projects</a></li>
<li><a href="#contact">Contact</a></li>
</ul>
</div>
</nav>
```

```
<!-- Hero Section -->
<section class="hero" id="home">
<canvas id="three-canvas"></canvas>
<div class="hero-content">
<h1 data-aos="fade-up">Vikram</h1>
<p data-aos="fade-up" data-aos-delay="200">Full-Stack Developer & Cloud Enthusiast</p>
<a href="#projects" class="cta-button" data-aos="fade-up" data-aos-delay="400">View My
Work</a>
</div>
</section>
```

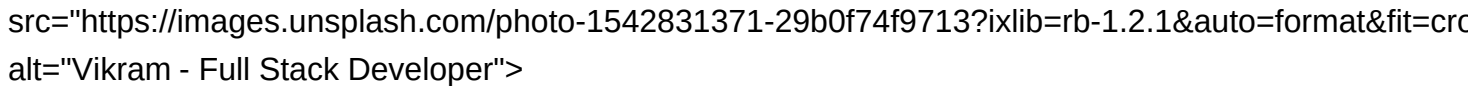
```
<!-- Ad Placeholder - Between Sections -->
<div class="container">
<div class="ad-container" data-aos="fade-up">
<!-- AdSense Ad Unit - Replace with your Ad Unit ID -->
<!-- <script async
src="https://pagead2.googlesyndication.com/pagead/js/adsbygoogle.js?client=YOURADUNIT
_ID" crossorigin="anonymous"></script> -->
<p>Advertisement Space</p>
</div>
</div>
```

```
<!-- About Section -->
<section id="about">
<div class="container">
<div class="section-title" data-aos="fade-up">
<h2>About Me</h2>
</div>
<div class="about-content">
<div class="about-text" data-aos="fade-right">
<p>I'm Vikram, a second-year BCA student with a strong interest in full-stack web
development and cloud computing.</p>
<p>I work with Python, Django, and SQL, and I enjoy building responsive web apps that solve
real problems.</p>
```

I'm curious about AI and love experimenting with new technologies. My goal is to join a company where I can grow as a developer and contribute to meaningful projects.

When I'm not coding, you'll find me exploring strategy games or reading about new tech trends.

[Get In Touch](#contact)



My Skills

Frontend Development

HTML5, CSS3, JavaScript, React, Responsive Design

Backend Development

Python, Django, SQL, REST APIs, Authentication

Cloud Computing

AWS, Azure, Docker, Serverless Architecture

AI & Machine Learning

TensorFlow, Scikit-learn, Data Analysis, NLP

```
</div>
</div>
</section>
```

```
<!-- Ad Placeholder - Between Sections -->
```

```
<div class="container">
```

```
<div class="ad-container" data-aos="fade-up">
```

```
<!-- AdSense Ad Unit - Replace with your Ad Unit ID -->
```

```
<!-- <script async
```

```
src="https://pagead2.googlesyndication.com/pagead/js/adsbygoogle.js?client=YOURADUNIT_ID" crossorigin="anonymous"></script> -->
```

```
<p>Advertisement Space</p>
```

```
</div>
```

```
</div>
```

```
<!-- Projects Section -->
```

```
<section id="projects">
```

```
<div class="container">
```

```
<div class="section-title" data-aos="fade-up">
```

```
<h2>My Projects</h2>
```

```
</div>
```

```
<div class="projects-container">
```

```
<div class="project-card" data-aos="fade-up" data-aos-delay="100">
```

```
<div class="project-image">
```

```

```

```
</div>
```

```
<div class="project-content">
```

```
<h3 class="project-title">E-commerce Platform</h3>
```

```
<div class="project-tech">
```

```
<span class="tech-tag">Django</span>
```

```
<span class="tech-tag">Python</span>
```

```
<span class="tech-tag">SQL</span>
```

```
<span class="tech-tag">JavaScript</span>
```

```
</div>
```

```
<p>A full-featured e-commerce platform with user authentication, product catalog, shopping
cart, and payment integration.</p>
```

```
</div>
```

```
</div>
```

```
<div class="project-card" data-aos="fade-up" data-aos-delay="200">
```

```
<div class="project-image">
```

```

</div>
<div class="project-content">
<h3 class="project-title">Task Management App</h3>
<div class="project-tech">
<span class="tech-tag">React</span>
<span class="tech-tag">Node.js</span>
<span class="tech-tag">MongoDB</span>
<span class="tech-tag">Express</span>
</div>
<p>A collaborative task management application with real-time updates, team collaboration,
and progress tracking.</p>
</div>
</div>
<div class="project-card" data-aos="fade-up" data-aos-delay="300">
<div class="project-image">

</div>
<div class="project-content">
<h3 class="project-title">Weather Dashboard</h3>
<div class="project-tech">
<span class="tech-tag">JavaScript</span>
<span class="tech-tag">API Integration</span>
<span class="tech-tag">Chart.js</span>
<span class="tech-tag">CSS3</span>
</div>
<p>An interactive weather dashboard that displays current conditions, forecasts, and historical
data with beautiful visualizations.</p>
</div>
</div>
</div>

<!-- Controls Section -->
<div class="controls-section" data-aos="fade-up">
<h3>Interactive Controls</h3>
<p>Adjust the settings below to see real-time changes in the 3D elements and media
compression:</p>
```



```
</div>
<div class="contact-container">
<div class="contact-info" data-aos="fade-right">
<h3>Let's Connect</h3>
<p>I'm always open to discussing new opportunities, collaborations, or just having a chat
about technology.</p>
```

```
<div class="contact-item">
<div class="contact-icon">📧 </div>
<div>
<h4>Email</h4>
<p>vikram873704@gmail.com</p>
</div>
</div>
```

```
<div class="contact-item">
<div class="contact-icon">📍 </div>
<div>
<h4>Location</h4>
<p>India</p>
</div>
</div>
```

```
<div class="contact-item">
<div class="contact-icon">🌐 </div>
<div>
<h4>LinkedIn</h4>
<p>https://www.linkedin.com/in/vikram-thapa-54485930a</p>
</div>
</div>
</div>
```

```
<div class="contact-form" data-aos="fade-left" data-aos-delay="200">
<form id="contactForm">
<div class="form-group">
<label class="form-label" for="name">Your Name</label>
<input type="text" id="name" class="form-input" required>
</div>
```

```
<div class="form-group">
<label class="form-label" for="email">Your Email</label>
<input type="email" id="email" class="form-input" required>
```

</div>

<div class="form-group">

<label class="form-label" for="subject">Subject</label>

<input type="text" id="subject" class="form-input" required>

</div>

<div class="form-group">

<label class="form-label" for="message">YourMessage</label>

<textarea id="message" class="form-textarea" required></textarea>

</div>

<button type="submit" class="cta-button">Send Message</button>

</form>

</div>

</div>

</div>

</section>

<!-- Footer -->

<footer>

<div class="container">

<div class="footer-content">

<div class="footer-column">

<h3>Vikram</h3>

<p>Full-Stack Developer & Cloud Enthusiast</p>

<div class="social-links">

in

gh

tw

ig

</div>

</div>

<div class="footer-column">

<h3>Quick Links</h3>

<ul class="footer-links">

Home

About

Skills

Projects

Contact


```
</ul>
</div>
```

```
<div class="footer-column">
<h3>Services</h3>
<ul class="footer-links">
<li><a href="#">Web Development</a></li>
<li><a href="#">Cloud Solutions</a></li>
<li><a href="#">API Development</a></li>
<li><a href="#">Consulting</a></li>
</ul>
</div>
</div>
```

```
<!-- Ad Placeholder - Footer -->
<div class="ad-container">
<!-- AdSense Ad Unit - Replace with your Ad Unit ID -->
<!-- <script async
src="https://pagead2.googlesyndication.com/pagead/js/adsbygoogle.js?client=YOUR_ADUNIT_ID" crossorigin="anonymous"></script> -->
<p>Advertisement Space</p>
</div>
```

```
<div class="copyright">
<p>&copy; 2025 Vikram. All Rights Reserved.</p>
</div>
</div>
</footer>
```

```
<!-- Ad Sidebar (Desktop Only) -->
<div class="ad-sidebar">
<!-- AdSense Ad Unit - Replace with your Ad Unit ID -->
<!-- <script async
src="https://pagead2.googlesyndication.com/pagead/js/adsbygoogle.js?client=YOUR_ADUNIT_ID" crossorigin="anonymous"></script> -->
<p>Advertisement</p>
</div>
```

```
<!-- Scripts -->
<script src="https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js"></script>
<script src="https://unpkg.com/aos@2.3.1/dist/aos.js"></script>
```

```
<script>
// Initialize AOS (Animate On Scroll)
document.addEventListener('DOMContentLoaded', function() {
  AOS.init({
    duration: 1000,
    once: true,
    offset: 100
  });

  // Mobile menu toggle
  const mobileMenuBtn = document.getElementById('mobileMenuBtn');
  const navLinks = document.getElementById('navLinks');

  mobileMenuBtn.addEventListener('click', function() {
    navLinks.classList.toggle('active');
  });

  // Navbar scroll effect
  window.addEventListener('scroll', function() {
    const navbar = document.getElementById('navbar');
    if (window.scrollY > 50) {
      navbar.classList.add('scrolled');
    } else {
      navbar.classList.remove('scrolled');
    }
  });

  // Slider controls
  const compressionSlider = document.getElementById('compressionSlider');
  const compressionValue = document.getElementById('compressionValue');

  const lightingSlider = document.getElementById('lightingSlider');
  const lightingValue = document.getElementById('lightingValue');

  const animationSlider = document.getElementById('animationSlider');
  const animationValue = document.getElementById('animationValue');

  const particleSlider = document.getElementById('particleSlider');
  const particleValue = document.getElementById('particleValue');

  compressionSlider.addEventListener('input', function() {
    compressionValue.textContent = this.value + '%';
  });
});
```

```
// In a real implementation, this would adjust image compression
updateVisuals();
});
```

```
lightingSlider.addEventListener('input', function() {
lightingValue.textContent = this.value + '%';
// In a real implementation, this would adjust lighting
updateVisuals();
});
```

```
animationSlider.addEventListener('input', function() {
animationValue.textContent = this.value + '%';
// In a real implementation, this would adjust animation speed
updateVisuals();
});
```

```
particleSlider.addEventListener('input', function() {
particleValue.textContent = this.value + '%';
// In a real implementation, this would adjust particle density
updateVisuals();
});
```

```
function updateVisuals() {
// This function would update the 3D scene based on slider values
// For demonstration, we'll just log the values
console.log('Compression:', compressionSlider.value);
console.log('Lighting:', lightingSlider.value);
console.log('Animation:', animationSlider.value);
console.log('Particles:', particleSlider.value);
}
```

```
// Three.js Scene
initThreeJS();
});
```

```
function initThreeJS() {
const canvas = document.getElementById('three-canvas');
const scene = new THREE.Scene();
const camera = new THREE.PerspectiveCamera(75, window.innerWidth /
window.innerHeight, 0.1, 1000);
const renderer = new THREE.WebGLRenderer({ canvas, alpha: true });
```

```
renderer.setSize(window.innerWidth, window.innerHeight);
renderer.setPixelRatio(Math.min(window.devicePixelRatio, 2));

// Create particles
const particlesGeometry = new THREE.BufferGeometry();
const particlesCount = 1000;

const posArray = new Float32Array(particlesCount * 3);

for (let i = 0; i < particlesCount * 3; i++) {
  posArray[i] = (Math.random() - 0.5) * 10;
}

particlesGeometry.setAttribute('position', new THREE.BufferAttribute(posArray, 3));

// Particle material
const particlesMaterial = new THREE.PointsMaterial({
  size: 0.02,
  color: 0x00f5ff,
  transparent: true,
  opacity: 0.8
});

const particlesMesh = new THREE.Points(particlesGeometry, particlesMaterial);
scene.add(particlesMesh);

// Add some geometric shapes
const geometry = new THREE.TorusGeometry(1, 0.4, 16, 100);
const material = new THREE.MeshBasicMaterial({
  color: 0x8a2be2,
  wireframe: true,
  transparent: true,
  opacity: 0.6
});
const torus = new THREE.Mesh(geometry, material);
scene.add(torus);

camera.position.z = 5;

// Animation
function animate() {
  requestAnimationFrame(animate);
```

```
particlesMesh.rotation.y += 0.001;
torus.rotation.x += 0.01;
torus.rotation.y += 0.005;

renderer.render(scene, camera);
}

animate();

// Handle window resize
window.addEventListener('resize', function() {
  camera.aspect = window.innerWidth / window.innerHeight;
  camera.updateProjectionMatrix();
  renderer.setSize(window.innerWidth, window.innerHeight);
});
}
</script>
</body>
</html>
```